

Software Developer Technical Test – Transition accelerator

Context:

We expect this test to take less than 3 hours to complete (around 1.5 hours for task 1 and 1.5 hours for task 2), and you're welcome to use AI tools such as Claude. Our goal is to get a sense of how you work and whether you'd be comfortable with the type of work this role involves. If you'd like a temporary Claude license, or have any questions about the tasks, don't hesitate to reach out.

Please submit your answers by **Wednesday, September 30th at 6pm EST**. All answers should be committed to this repo: <https://github.com/The-Transition-Accelerator/developer-exam.git>

Task 1.1

Using the attached data (transportation_co2_scenarios.csv and transportation_baseline_indicators.csv, extracted from the EIA Annual Energy Outlook 2026), build an interactive, public-facing dashboard that lets a user explore an additional transportation-electrification scenario layered on top of the Counterfactual Baseline case.

What to build: A single HTML file (with embedded JS — no server, no build step, no external dependencies beyond what can be loaded via public CDN) that:

1. Plots the Counterfactual Baseline and Alternative Transportation emissions paths (2025–2050) from transportation_co2_scenarios.csv as reference lines.
2. Lets the user tune two assumptions and see a third line update live:
 - a. % of light-duty vehicle miles electrified, on top of whatever electrification is already implicit in the baseline (0–100%, slider or input)
 - b. One assumption of your choosing that affects how much emissions each electrified mile avoids (e.g., grid emissions intensity, an EV efficiency figure, or something else you think matters) — you decide what it is and where the number comes from.
3. Recomputes and redraws the custom-scenario emissions line as those inputs change.

You must also state, in a short README or comment block:

1. The one assumption you chose in (2) above, its value, and where it came from (a source, a reasonable estimate, or your own back-of-envelope reasoning — just be explicit).
2. The formula you used to go from "% VMT electrified" + your assumption to an emissions number.
3. Anything you'd want to double-check or refine with more time.

What we're looking for: we care less about whether your number is "right" and more about whether your assumptions are legible, your formula is defensible, and the dashboard is something a non-technical person could use without an explanation.

Task 1.2:

Explain how you would estimate the impact of the additional vehicle electrification scenario in task 1.1 on electricity generation. Describe the main assumptions and data you would need. You do not need to submit code for this answer, plain text in a .md file explaining your reasoning is enough.

Task 2:

The Transition Accelerator has a wealth of institutional knowledge about energy policies and measures that is used as assumptions for modelling by different individuals across multiple teams. Could you suggest an architecture for a solution that would let multiple models refer to these assumptions, ensure traceability of sources, allow for easy updates of values when necessary but also allow for users to conduct sensitivity analyses when appropriate? Please provide a diagram of the different systems you see interacting with each other, how they interact (event-based or continuous – in what format), and how they are secured. Feel free to add written notes explaining your reasoning, and include any questions you might ask your future colleagues to help craft an appropriate solution.